| | UNIT III – TESTING OF HYPOTHESIS |
|---|---|
| | Sampling distributions - Estimation of parameters - Statistical hypothesis - Large sample tests based on Normal distribution for single mean and difference of means -Tests based on t, Chi-square and F distributions for mean, variance and proportion - Contingency table (test for independent) - Goodness of fit. |
| | **PART *A** |
| **Q.No.** | **Questions** |
| 1. | **Mention the various steps involved in testing of hypothesis. (A.U.A/M 2010)** BTL1<br>(i) Set up a null hypothesis H0.<br>(ii) Set up the alternative hypothesis H1.<br>(iii) Select the appropriate level of significance (α).<br>(iv) Compute the test statistic under H0<br>(v) We compare the "calculate value" with "critical value " at given level of significance (α). |
| 2 | **Define Chi–Square test for goodness of fit. (A.U.A/M 2010)** BTL1<br>Chi–Square test of goodness of fit is a test to find if the deviation of the experiment from theory is just by chance or it is due to the inadequacy of the theory to fit the observed data. By this test, we test whether difference between observed and expected frequencies are significant or not.<br>Chi–Square test statistic of goodness of fit is defined by $\psi^2 = \sum \frac{(O-E)^2}{E}$, where O is the observed frequency and E expected frequency. |
| 3 | **Give the main use of ψ2 –test? (A.U.N/D 2010, 2011)** BTL2<br>1. To test the significance of discrepancy between the experimental values and the theoretical values, obtained under some theory or hypothesis.<br>2. With the help of **ψ2 –test,** we can find out whether two or more attributes are associated or not. |
| 4 | **Write two applications of $\psi^2$ test. (A.U.N/D 2010, M/J 2012)** BTL3<br>(i) To test the goodness of fit<br>(ii) To test the independent of attributes<br>(iii) To test the homogeneous of independent estimations. |
| 5 | **What are the applications of *t*-distributions? (A.U.A/M 2011)** BTL3<br>(i) To test the significance of a single mean.<br>(ii) To test the significance of the difference between two sample means.<br>(iii) To test the significance of the coefficient of correlation. |
| 6 | **It has been found that 2% of the tools produced by a certain machine are defective. What is probability that in a shipment of 400 such tools 3% are more?. (A.U.A/M 2011,M/J 2012,N/D 2013, M/J 2014)** BTL1<br>YP=0.0007, ans=0.1056 |

| | |
|---|---|
| | Type II error : Accept H0 when it is wrong. |
| 7 | **Define Null and Alternate Hypothesis. (A.U. M/J 2012)** BTL1<br>For applying the test of significance, we first set up of a hypothesis a definite statement about the population parameter. Such a hypothesis is usually a hypothesis of no difference and it is denoted by H0.<br>Any hypothesis which is complementary to the null hypothesis **(H0)** is called an alternative hypothesis, denoted by H1. |
| 8 | **Define Level of significance. (A.U M/J 2011,A.U.N/D 2013)** BTL1<br>The probability that the value of the statistic lies in the critical region is called the level of significance.<br>In general, these levels are chosen as 0.01 and 0.05, called 1% level and 5% level of significance respectively. |
| 9 | **State the conditions for applying ψ2 –test . (A.U.M/J 2014)** BTL1<br>(i) The sample observations should be independent.<br>(ii) Constraints on the cell frequencies, if any, must be linear.<br>(iii) N, the total frequency should be at least 50.<br>(iv) No. of theoretical cell frequency should be less than 5. |
| 10 | **Give the formula for the ψ2 –test of independence for** $\boxed{\begin{matrix} a & b \\ c & d \end{matrix}}$ **(A.U.N/D 2012,N/D2014)** BTL3<br>Formula is $\dfrac{(a+b+c+d)(ad-bc)^2}{(a+c)(b+d)(a+b)(c+d)}$ |
| 11 | **Define Error. (A.U.N/D 2009)** BTL1<br>In sampling theory to draw valid inferences about the population parameter on the basis of the sample results we decide to accept or to reject the H0 after examining a sample from it.<br>Type I error : Reject H0 when it is true.<br>Type II error : Accept H0 when it is wrong. |
| 12 | **The heights of college students in Chennai are normally distributed with S.D 6cm and sample of 100 students had their mean height 158cm. Test the hypothesis that the mean height of college students in Chennai is 160cm at 1% level of significance. (A.U.N/D 2011)** BTL5<br>$H_0 : \mu = 160;\ H_1 : \mu \neq 160$<br>$Z = \dfrac{X-\mu}{\frac{s}{\sqrt{n-1}}} = \dfrac{158-160}{\frac{6}{\sqrt{100}}} = -3.33 > 2.58$<br>$H_0$ is rejected. |
| 13 | **What is sampling distribution? [AU N/D 2018]**<br>The frequency distribution so formed is known as sampling distribution of the mean, similarly, sampling distribution of standard deviation we can have. |
| 14 | **What are the parameters and statistics in sampling? [AU N/D 2011]**<br>Parameters $\{\mu - mean,\ \sigma - Standared\ deviation\}$ of population<br>Statistics $\{\bar{x} - mean,\ s - Standared\ deviation\}$ of sample |
| 15 | **Mention the various steps involved in testing of hypothesis. [AU N/D 2010]** |

| | |
|---|---|
| | i. Set up a null hypothesis $H_0$,<br>ii. Set up the alternative hypothesis $H_1$,<br>iii. Select the appropriate level of significance $(\alpha)$,<br>iv. Compute the test statistics z under $H_0$,<br>v. Compare the calculate z with critical value $z_\alpha$ at given level of $(\alpha)$, |
| 16 | **Write down the formula of test statistic 't' to test the significance of the difference between the means. [AU N/D 2016; A/M 2015]**<br>$Z = \left\| \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \right\|$ ; $\overline{x_1}, \overline{x_2}$ sample means; $s_1^2, s_2^2$ sample variances; $n_1, n_2$ sample sizes |
| 17 | **Define students t-test for difference of means of two samples. [AU N/D 2010]**<br>$t = \frac{\overline{x_1} - \overline{x_2}}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ with $n_1 + n_2 - 2$<br>Where $s^2 = \frac{\sum(x_1 - \overline{x_1})^2 + \sum(x_2 - \overline{x_2})^2}{n_1 + n_2 - 2}$ |
| 18 | **What is Type-I, Type 2 errors? [AU M/J 2014, N/D 2013, A/M 2011, M/J 2016**<br>Type-I Error: Rejecting $H_0$ when it is true.<br>Type-II Error: Accepting $H_0$ when it is false. |
| 19 | **It has been found that 2% of the tools produced by a certain machine are defective. What is probability that in a shipment of 400 such tools 2% or less than will prove defective?**<br>UP=0.02; Required probability=0.5714 |
| 20 | **For the following cases, specify which probability distribution to use in a hypothesis test.**<br>**a) $H_0: \mu = 27, H_1: \mu \neq 27, \overline{X} = 20.1, \sigma = 5, n = 12$**<br>**b) $H_0: \mu = 98.6, H_1: \mu > 98.6, \overline{X} = 65, \sigma = 12, n = 12$**<br>(a) Normal distribution: since $\sigma$ is given.<br>(b) t-distributed with d.f = 41. |
| | **Part*B** |
| 1 | **A sample of 900 members has a mean 3.4cm and standard deviation 2.61cm. Is the sample from a large population of mean 3.25cms and standard deviation of 2.61cms? (8M) (A.U.A/M 2010)** BTL2<br>**Answer: Page : 3.17 – Dr.A. Singaravelu**<br>$H_0 : \mu = 3.25;\ H_1 : \mu \neq 3.25$ (2M)<br>$Z = \frac{X - \mu}{\frac{s}{\sqrt{n-1}}} = \frac{3.4 - 3.25}{\frac{2.61}{\sqrt{900}}} = 1.724 > 1.96$ (4M)<br>Accept $H_0$. (2M) |
| 2 | **The means of 2 large samples 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples the regarded as drawn from the same population of SD 2.5inches. (8M)** BTL4<br>**Answer: Page : 3.19 – Dr.A. Singaravelu**<br>$H_0 : \mu_0 = \mu_1;\ H_1 : \mu_0 \neq \mu_1$ (2M) |

$$Z = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} = \frac{67.5 - 68}{\sqrt{(2.5)^2\left(\frac{1}{1000} + \frac{1}{2000}\right)}} = -5.16 > 1.96 \quad (4M)$$

$H_0$ is rejected at 5% level of significance. (2M)

3

**A machinist is making engine parts with axle diameters of 0.7 inch. A random sample of 10 parts shows a mean diameter if 0.742 inch with a S.D. of 0.040 inch. Compute the statistic you would used to test whether the work is meeting the specification. (8M)** BTL3

**Answer: Page : 3.25 – Dr.A. Singaravelu**

$H_0 : \mu_0 = \mu_1;\ H_1 : \mu_0 \neq \mu_1$ (2M)

$$t = \frac{X - \mu}{\frac{s}{\sqrt{n-1}}} = \frac{0.742 - 0.7}{\frac{0.040}{\sqrt{9}}} = 3.15 > 2.26 \quad (4M)$$

$H_0$ is rejected at 5% level of significance. (2M)

4

**The average number of articles to be produced by 2 machine per day are 200 and 250 with S.D of 20 and 25 respectively on the basis of record, of 25 days production can you regard the machines equally efficient at 1% level of significance. (8M)** BTL4

**Answer: Page : 3.38 – Dr.A. Singaravelu**

$H_0 : \mu_0 = \mu_1;\ H_1 : \mu_0 \neq \mu_1$ (2M)

$$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = 7.65 > 2.58 \quad (4M)$$

$H_0$ is rejected at 1% level of significance. (2M)

5

**The following data gives the number of aircraft accidents that occurred during the various days of a week Find whether the accidents are uniformly distributed over the week. (A.U.N/D 2010) (8M)**
BTL3

| **Day :** | **SUN** | **MON** | **TUE** | **WED** | **THUR** | **FRI** | **SAT** |
|---|---|---|---|---|---|---|---|
| **No. of accidents:** | **14** | **16** | **8** | **12** | **11** | **9** | **14** |

**Answer: Page : 3.61 – Dr.A. Singaravelu**

$H_0$**:** The accidents are uniformly distributed over the week. (2M)

| Observed Frequency | Expected Frequency | $(O-E)^2$ | $\frac{(O-E)^2}{E}$ |
|---|---|---|---|
| 14 | 14 | 0 | 0 |
| 18 | 14 | 16 | 1.143 |
| 12 | 14 | 4 | 0.286 |
| 11 | 14 | 9 | 0.643 |

| | | | |
|---|---|---|---|
| 15<br>14 | 14<br>14 | 1<br>0 | 0.071<br>0 |
| | | | 2.143 |

$$\psi^2 = \sum \frac{(O-E)^2}{E} = 2.143 < 11.07 \qquad (4M)$$

Accept $H_0$, the accidents are uniformly distributed over the week. (2M)

6

**Theory predicts that the proportion of beans in four groups A, B, C, D should be 9:3:3:1. In an experiment among 1600 beans, the numbers in the four groups were 882, 313, 287 and 118. Does the experiment support the theory? (A.U.M/J 2012) (8M)** BTL5

**Answer: Page : 3.62 – Dr.A. Singaravelu**

$H_0$ **:** The experimental result support the theory. (2M)

| Observed Frequency | Expected Frequency | $(O-E)^2$ | $\frac{(O-E)^2}{E}$ |
|---|---|---|---|
| 882<br>313<br>287<br>118 | 900<br>300<br>300<br>100 | 324<br>169<br>169<br>324 | 0.360<br>0.563<br>0.563<br>3.240 |
| | | | 4.726 |

$$\psi^2 = \sum \frac{(O-E)^2}{E} = 4.726 < 7.81 \qquad (4M)$$

Accept $H_0$, the experimental results supports the theory. (2M)

7

**A soap manufacturing company was distributing a particular brand of soap through a large no of retail shops. Before a heavy advertisement campaign, the mean sales per week per shop were 146.3 dozens. After the campaign, a sample of 22 shops was taken and the mean sales were found to be 153.7 dozen with SD =17.2. Can you consider the advertisement effective? (8M)** BTL6

**Answer: Page : 3.25 – Dr.A. Singaravelu**

$H_0 : \mu_0 = \mu_1;\ H_1 : \mu_0 > \mu_1\ \left(One\ Tail\right)$ (2M)

$$t = \frac{X-\mu}{\frac{s}{\sqrt{n-1}}} = \frac{153.7-146.3}{\frac{17.2}{\sqrt{21}}} = 1.97 > 1.72 \qquad (4M)$$

$H_0$ is rejected at 5% level of significance. (2M)

8

**The following table gives the classification of 100 workers according to sex and nature of work. Test whether the nature of work is independent of the sex of the worker. (8M)** BTL4

| | **Stable** | **Unstable** | **Total** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **Males** | **40** | **20** | **60** |
| **Females** | **10** | **30** | **40** |
| | **50** | **50** | **100** |

**Answer: Page : 3.65 – Dr.A. Singaravelu**

$H_0$ : the nature of work is independent of the sex of the workers. (2M)

| Observed Frequency | Expected Frequency | $(O-E)^2$ | $\frac{(O-E)^2}{E}$ |
|---|---|---|---|
| 40 | 30 | 100 | 3.33 |
| 20 | 30 | 100 | 3.33 |
| 10 | 20 | 100 | 5 |
| 30 | 20 | 100 | 5 |
| | | | 16.66 |

$$\psi^2 = \sum \frac{(O-E)^2}{E} = 16.66 < 3.84$$ (4M)

Accept $H_0$, there is difference in the nature of work on the basis of sex. (2M)

9

**Given the following contingency table for hair colour and eye colour. Find the value of $\psi^2$. Is there good association between the two.** **(8M)** BTL5

| | | **Hair Colour** | | | **Total** |
|---|---|---|---|---|---|
| | | **Fair** | **Brown** | **Black** | |
| **Eye Colour** | **Blue** | **15** | **5** | **20** | **40** |
| | **Grey** | **20** | **10** | **20** | **50** |
| | **Brown** | **25** | **15** | **20** | **60** |
| | **Total** | **60** | **30** | **60** | **150** |

**Answer: Page : 3.69 – Dr.A. Singaravelu**

$H_0$ : The two attributes Hair colour and Eye colour are independent (2M)

| Observed Frequency | Expected Frequency | $(O-E)^2$ | $\frac{(O-E)^2}{E}$ |
|---|---|---|---|
| 15 | 16 | 1 | 0.0625 |
| 5 | 8 | 9 | 1.125 |

| | | | | | |
|---|---|---|---|---|---|
| | 20 | 16 | 16 | 1 | |
| | 20 | 20 | 0 | 0 | |
| | 10 | 10 | 0 | 0 | |
| | 20 | 20 | 0 | 0 | |
| | 25 | 24 | 1 | 0.042 | |
| | 15 | 12 | 9 | 0.75 | |
| | 20 | 24 | 16 | 0.66 | |

$$\psi^2 = \sum \frac{(O-E)^2}{E} = 3.6458 < 9.488 \qquad (4M)$$

Accept $H_0$, The hair colour and eye colour are independent. (2M)

10 **Test whether there is any significant difference between the variance of the population from which the following samples are taken: (A.U.N/D 2012) (8M)** BTL3

| |
|---|
| **Samples I : 20 16 26 27 23 22** |
| **Samples II: 27 33 42 35 32 34 38** |

**Answer: Page : 3.53 – Dr.A. Singaravelu**

$\overline{x}_1 = 22.3$ $\quad \overline{x}_2 = 34.4$

$s_1^2 = 16.268$ $\quad s_2^2 = 22.29$ (2M)

$H_0$: There is no significant difference between the variances. (1M)

$F = \frac{S_2^2}{S_1^2} = 1.37 < 4.95$ (4M) Accept $H_0$. (1M)

## UNIT VI DESIGN OF EXPERIMENTS

One way and Two way classifications - Completely randomized design – Randomized block design – Latin square design

### PART *A

| Q.No. | Questions |
|---|---|
| 1. | **Discuss the advantages and disadvantages of Randomized block design. (A.U.A/M 2010)** BTL2<br>**Advantage:**<br>(i) Analysis is possible, even if some observations are missing.<br>(ii) The analysis of the design is sample as it result in a two-way classification analysis of variance.<br>**Disadvantage:**<br>(i) If the interactions are large, the experiment may yield misleading results.<br>(ii) The shape of the experimental material should be rectangular. |
| 2 | **State the advantages of a factorial Experiment over a simple experiment. (A.U.A/M 2010, 2014)** BTL2<br>Factorial experiment is the procedure of varying all factors simultaneously. |

| | |
|---|---|
| | A major conceptual advancement in experimental design is exemplified by factorial design.<br>In factorial designs, an assessment of each individual factor effect is based on the whole set of measurements so that a more efficient utilization of experimental resources is achieved in the these designs. |
| 3 | **Compare one-way classification model with two-way classification model. (A.U.N/D 2010, 2011)** BTL2<br>**One way Classification** — **Two way classification**<br>One factor is involved — Two factors involved<br>One set of hypothesis — Two set of hypothesis |
| 4 | **What is meant by Latin square? (A.U.N/D 2010)** BTL3<br>The Latin Square model assumes that interactions between treatments and row and column groupings non-existent. Since each treatment occurs only once in each row or column, if interactions are present, it is possible for them to cause an apparently significant difference between treatment. |
| 5 | **What are the applications of *t*-distributions? (A.U.A/M 2011)** BTL2<br>(i) To test the significance of a single mean.<br>(ii) To test the significance of the difference between two sample means.<br>(iii) To test the significance of the coefficient of correlation. |
| 6 | **State the assumptions involved in ANOVA. (A.U.M/J 2012)** BTL1<br>(i). Normality<br>(ii). Homogeneity<br>(iii). Independence of error. |
| 7 | **What are the advantages of a Latin square design? (A.U.M/J 2012)** BTL2<br>(i) The analysis is simple, it is only slightly more complicated than that for the randomized complete block design.<br>(ii) The analysis remains relatively simple even with missing data. Analytical procedures are available for omitting one or more treatment, rows or columns. |
| 8 | **State the basic principles of design of Experiments. (A.U.N/D 2012), (A.U.M/J 2014)** BTL2<br>The basic principles of the design of experiments are<br>(i) Replication<br>(ii) Randomization<br>(iii) Local control. |
| 9 | **Define :RBD** BTL1<br>Let us consider an agricultural experiment using which we wish to test the effect of „k" fertilising treatment on the yield of crops. We assume that we know some information about the soil fertility of the plots. Then we divide the plots into „h" blocks, according to the soil fertility each block containing „k" blocks.<br>Thus the plots in each block will be of homogeneous fertility as far as possible within each block, the „k" treatments are given to the „k" plots in a perfectly random manner, such that each treatment occurs only once in any block. But the same k treatment are from block to block. This design is called **Randomized Block Design.** |
| 10 | **What do you understand by "Design of Experiments"? (A.U.A/M 2013)** BTL2<br>The design of experiment may be defined as the logical construction of the experiment in which the degree of uncertainty with which the inference is drawn may be well defined. |
| 11 | **Define types of Error. (A.U.N/D 2009)** BTL1 |

In sampling theory to draw valid inferences about the population parameter on the basis of the sample results we decide to accept or to reject the H0 after examining a sample from it.

Type I error : Reject H0 when it is true.

Type II error : Accept H0 when it is wrong.

12 **Write down the ANOVA table for one way classification. (A.U.A/M 2013)** BTL4

Analysis of Variance:

| Source of Variation | SS(Sum of Squares) | Degree of Freedom | Mean Square | Variance Ratio of F |
|---|---|---|---|---|
| Between Sample | SSC | $V_1 = c - 1$ | $MSC = \frac{SSC}{(c-1)}$ | $F_C = \frac{MSC}{MSE}$ |
| Within Sample | SSE | $V_2 = n - c$ | $MSE = \frac{SSE}{(n-c)}$ | |
| Total | TSS | n- 1 | | |

13 **Define $2^2$ factorial design. (A.U.N/D 2013)** BTL1

A complete replicate of such a design requires $2 \times 2 \times \ldots\ldots \times 2 = 2^k$ observations is called a $2^k$ factorial design $2^2$ is the simplest design $2^k$.

i.e., Two factors A and B each at two level that it may be low and high levels of the factor.

14 **State any two advantages of a CRD. (A.U.M/J 2014)** BTL2

(i) It has a simple lay out.

(ii) There is complete flexibility as the number of replication is not fixed.

(iii) Analysis can be performed, if some observation are missing.

15 **When do you apply analysis of variance technique? (A.U.M/J 2005)**

To test the homogenity of several means.

16 **Define the term completely randomized design. (A.U.M/J 2005)**

The completely randomized design is the simplest of all the designs, based on principles of randomization and replication. In this design, treatments are allocated at random to the experimental units over the entire experimental material.

17 **Define mean square. (A.U.M/J 2007)**

Mean square = Sum of the squares/Degrees of freedom

18 **What do you mean by Replication? (A.U.M/J 2007)**

Replication means the repetition of the treatments under investigation. An experimenter resorts to replication to average out the influence of the chance factors on different experimental units.

19 **What do you mean by two-way classification in analysis of variance?**

In a two-way classification the data are classified according to two different criteria or factors.

20 **What is a randomized block design? (A.U.M/J 2005)**

Let us consider an agricultural experiment using which we wish to test the effect of 'k' fertilizing treatments on the yield of crops. We assume that we know some information about the soil fertility of the plots. Then we divide the plots into 'h' blocks, according to the soil fertility each block containing 'k' blocks. Thus the plots in each block will be of homogeneous fertility as far as possible within each block, the ‘k’ treatments are given to the ‘k’ plots in a perfectly random manner, such that each treatment occurs only once in any block. But the same k treatments are repeated from block to block.

**Part*B**

1

**Three different machines are used for a production. On the basis of the outputs, set up one-way ANOVA table and test whether the machines are equally effective.**

| **Outputs** | | |
|---|---|---|
| **Machine I** | **Machine II** | **Machine III** |
| **10** | **9** | **20** |
| **15** | **7** | **16** |
| **11** | **5** | **10** |
| **10** | **6** | **14** |

**Given that the value of F at 5% level of significance for (2, 9) is 4.26.** **(16M)** **BTL3**

**Answer: Page : 4.10 – Dr.A. Singaravelu**

$H_0$: The machines are equally effective

1. $H_0$ 2. $H_1$ 3. Find the correction factor $\frac{T^2}{N}$ =1704.08 (2M)

**4.** $TSS = \sum X_1^2 + \sum X_2^2 + \cdots - \frac{T^2}{N}$ =284.92 (2M)

**5.** $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_2} + \frac{(\sum X_3)^2}{N_3} \cdots - \frac{T^2}{N}$ =162.17 (2M)

**6.** $SSE = TSS - SSC$ =122.75 (2M)

7. Prepare the ANOVA table to calculate F-ratio, it should be calculated in such a way that $F_C > 1$ .
8. Conclusion (6M)

| Source of variation | Sum of squares | d.f. | Mean square | Variance ratio | Table value at $\alpha$ % level |
|---|---|---|---|---|---|
| Between columns | SSC=162.17 | $C-1$ =2 | $MSC = \frac{SSC}{C-1}$ =81.085 | $F_C = \frac{MSC}{MSE}$ =5.95 | 4.26 |
| Error | SSE=122.75 | $N-C$ =9 | $MSE = \frac{SSE}{N-C}$ =13.63 | | |

Reject the hull hypothesis $H_0$. That is the three machines are not equally effective. (2M)

2

**The following data represent the number of units of production per day turned out by 5 different workers using 4 different types of machines : (A.U.N/D 2011) (A.U.A/M 2013) (16M)** BTL4

**Machine type**

| | | A | B | C | D |
|---|---|---|---|---|---|
| | 1 | 44 | 38 | 47 | 36 |
| **Workers** | 2 | 46 | 40 | 52 | 43 |
| | 3 | 34 | 36 | 44 | 32 |
| | 4 | 43 | 38 | 46 | 33 |
| | 5 | 38 | 42 | 49 | 39 |

**(a) Test whether the mean production is the some for the different machine type.**

**(b) Test whether the 5 mean differ with mean productivity.**

**Answer: Page : 4.30 – Dr.A. Singaravelu**

Code the data by subtracting 40 from each value.

1. $H_0$ 2. $H_1$ 3. $C.F. = \frac{T^2}{N}$ =20 4. $TSS = \sum X_1^2 + \sum X_2^2 + \cdots - \frac{T^2}{N}$ =574

5. $SSC = \frac{\left(\sum X_1\right)^2}{N_1} + \frac{\left(\sum X_2\right)^2}{N_2} + \frac{\left(\sum X_3\right)^2}{N_3} \cdots - \frac{T^2}{N}$ =338.8

6. $SSR = \frac{\left(\sum Y_1\right)^2}{N_1'} + \frac{\left(\sum Y_2\right)^2}{N_2'} + \frac{\left(\sum Y_3\right)^2}{N_3'} \cdots - \frac{T^2}{N}$ =161.5

7. $SSE = TSS - SSC - SSR$ =73.7 8. Prepare ANOVA table for two way classification:

| Source of variation | Sum of squares | d.f. | Mean square | Variance ratio | Table value at $\alpha$ % level |
|---|---|---|---|---|---|
| Between columns | $SSC$ =338.8 | $C-1$ =3 | $MSC = \frac{SSC}{C-1}$ =112.93 | $F_C = \frac{MSC}{MSE}$ =18.38 | 3.49 |
| Between rows | $SSR$ =161.5 | $R-1$ =4 | $MSR = \frac{SSR}{R-1}$ =40.37 | $F_R = \frac{MSR}{MSE}$ =6.574 | 3.26 |
| Error | $SSE$ =73.7 | $(C-1).(R-1)$ =12 | $MSE = \frac{SSE}{(R-1)(C-1)}$ =6.142 | | |

9. Conclusion

The workers differ with respect to mean productivity.

3

**A variable trial was conducted on wheat with 4 varieties in a Latin square design. The plan of the experiment and the per plot yield are given below. (16M) (A.U.N/D 2011) BTL4**

| | | | |
|---|---|---|---|
| **C25** | **B23** | **A20** | **D20** |
| **A19** | **D19** | **C21** | **B18** |
| **B19** | **A14** | **D17** | **C20** |
| **D17** | **C20** | **B21** | **A15** |

**Answer: Page : 4.46 – Dr.A. Singaravelu**

Code the data by subtracting 20 from each value.

1. $H_0$ 2. $H_1$ 3. $C.F.=\frac{T^2}{N}$ =9 (2M)4. $TSS=\sum X_1^2+\sum X_2^2+\cdots-\frac{T^2}{N}$ =113 (2M)

5. $SSC=\frac{\left(\sum X_1\right)^2}{N_1}+\frac{\left(\sum X_2\right)^2}{N_1}+\frac{\left(\sum X_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =7.5 (2M)

6. $SSR=\frac{\left(\sum Y_1\right)^2}{N_1}+\frac{\left(\sum Y_2\right)^2}{N_1}+\frac{\left(\sum Y_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =46.5 (2M)

7. $SSK=\frac{\left(\sum Z_1\right)^2}{N_1}+\frac{\left(\sum Z_2\right)^2}{N_1}+\frac{\left(\sum Z_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =48.5 (2M)

8. $SSE=TSS-SSC-SSR-SSK$ =10.5

9. ANOVA table: (4M)

| **Source of variation** | **Sum of squares** | **d.f.** | **Mean square** | **Variance ratio** | **Table value at 5% level** |
|---|---|---|---|---|---|
| Between columns | $SSC$ =7.5 | 3 | $MSC=\frac{SSC}{K-1}$ =2.5 | $F_C=\frac{MSC}{MSE}$ =1.429 | 4.76 |
| Between rows | $SSR$ =46.5 | 3 | $MSR=\frac{SSR}{K-1}$ =15.5 | $F_R=\frac{MSR}{MSE}$ =8.85 | 4.76 |
| Between treatment | $SSK$ =48.5 | 3 | $MSK=\frac{SSK}{K-1}$ =16.17 | $F_K=\frac{MSK}{MSE}$ =9.24 | 4.76 |
| Error | $SSE$ =10.5 | 6 | MSE=1.75 | | |

10. Conclusion (2M)

Row-varieties and treatments-varieties are rejected and Column-yields is accepted.

4

**A farmer wishes to test the effect of 4 fertilizers A,B,C,D on the yield of wheat. The fertilizers are used in a LSD and the result are tabulated here. Perform an analysis of variance. (A.U.N/D 2012) (16M)** BTL5

| | | | |
|---|---|---|---|
| **A18** | **C21** | **D25** | **B11** |
| **D22** | **B12** | **A15** | **C19** |
| **B15** | **A20** | **C23** | **D24** |
| **C22** | **D21** | **B10** | **A14** |

**Answer: Page : 4.60 – Dr.A. Singaravelu**

Code the data by subtracting 20 from each value.

1. $H_0$ 2. $H_1$ 3. $C.F.=\frac{T^2}{N}$ =14.06 (2M) 4. $TSS=\sum X_1^2+\sum X_2^2+\cdots-\frac{T^2}{N}$ =354.94 (2M)

| | |
|---|---|
| | 5. $SSC=\frac{\left(\sum X_1\right)^2}{N_1}+\frac{\left(\sum X_2\right)^2}{N_1}+\frac{\left(\sum X_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =19.69 (2M)<br>6. $SSR=\frac{\left(\sum Y_1\right)^2}{N_1}+\frac{\left(\sum Y_2\right)^2}{N_1}+\frac{\left(\sum Y_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =19.19 (2M)<br>7. $SSK=\frac{\left(\sum Z_1\right)^2}{N_1}+\frac{\left(\sum Z_2\right)^2}{N_1}+\frac{\left(\sum Z_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =288.19 (2M)<br>8. $SSE=TSS-SSC-SSR-SSK$ =27.87 (2M)<br>9. ANOVA table: (4M) |

| **Source of variation** | **Sum of squares** | **d.f.** | **Mean square** | **Variance ratio** | **Table value at 5% level** |
|---|---|---|---|---|---|
| Between columns | $SSC$ =19.69 | 3 | $MSC=\frac{SSC}{K-1}$ =6.397 | $F_C=\frac{MSC}{MSE}$ =1.413 | 4.76 |
| Between rows | $SSR$ =19.19 | 3 | $MSR=\frac{SSR}{K-1}$ =6.563 | $F_R=\frac{MSR}{MSE}$ =1.377 | 4.76 |
| Between treatment | $SSK$ =288.19 | 3 | $MSK=\frac{SSK}{K-1}$ =96.06 | $F_K=\frac{MSK}{MSE}$ =20.68 | 4.76 |
| Error | $SSE$ =27.87 | 6 | MSE=4.645 | | |

11. Conclusion (2M)

Row and column-varieties are accepted treatment-yields is rejected.

5

**An experiment was designed to study the performance of 4 different detergents for cleaning fuel injectors. The following cleanliness readings were obtained with specially designed equipment for 12 tanks of gas distributed over 3 different models of engines.**

| **Seasons** | **Salesmens** | | | | **Total** |
|---|---|---|---|---|---|
| | **A** | **B** | **C** | **D** | |
| **Summer** | **36** | **36** | **21** | **35** | **128** |
| **Winter** | **28** | **29** | **31** | **32** | **120** |
| **Monsoon** | **26** | **28** | **29** | **29** | **112** |
| **Salesmens Total** | **90** | **93** | **81** | **96** | **360** |

i) **Do the salesmen significantly differ in performance?**

ii) **Is there significant difference between the seasons?**

**[A.U. A/M 2008, A.U N/D 2011, A.U Tvil M/J, A.U A/M 2015]**

**Answer : Dr.Balaji [P&S] Edition 1, page no : 4.32.**

**Step 1: N=12**

**Step 2: T=-0**

**Step 3: C.F=0**

| | |
|---|---|
| | **Step 4: TSS=210**<br>**Step 5: SSC=42**<br>**Step 6: SSR=32**<br>**SSE=136**<br>**Step 7: Table of analysis of variance**<br>**Step 8: Conclusion: No significant difference.** |
| 6 | **Analyse the variance in the Latin square of yields( in kgs) of paddy where P,Q,R,S denote the different methods of cultivation.**<br>**S122 P121 R123 Q122**<br>**Q124 R123 P122 S125**<br>**P120 Q119 S120 R121**<br>**R122 S123 Q121 P121**<br>**Examine whether different method of cultivation have significantly different yields.(16M) (A.UA/M 2014)** BTL3 |

**Answer: Page : 4.55 – Dr.A. Singaravelu**

Code the data by subtracting 120 from each value.

1. $H_0$ 2. $H_1$ 3. $C.F.=\frac{T^2}{N}$ =56.25 (2M) 4. $TSS=\sum X_1^2+\sum X_2^2+\cdots-\frac{T^2}{N}$ =35.75 (2M)

5. $SSC=\frac{\left(\sum X_1\right)^2}{N_1}+\frac{\left(\sum X_2\right)^2}{N_1}+\frac{\left(\sum X_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =2.75 (2M)

6. $SSR=\frac{\left(\sum Y_1\right)^2}{N_1}+\frac{\left(\sum Y_2\right)^2}{N_1}+\frac{\left(\sum Y_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =24.75 (2M)

7. $SSK=\frac{\left(\sum Z_1\right)^2}{N_1}+\frac{\left(\sum Z_2\right)^2}{N_1}+\frac{\left(\sum Z_3\right)^2}{N_1}\cdots-\frac{T^2}{N}$ =4.25 (2M)

8. $SSE=TSS-SSC-SSR-SSK$ =4 (2M)

9. ANOVA table: (2M)

| **Source of variation** | **Sum of squares** | **d.f.** | **Mean square** | **Variance ratio** | **Table value at 5% level** |
|---|---|---|---|---|---|
| Between columns | $SSC$ =2.75 | 3 | $MSC=\frac{SSC}{K-1}$ =8.25 | $F_C=\frac{MSC}{MSE}$ =1.369 | 4.76 |
| Between rows | $SSR$ =24.75 | 3 | $MSR=\frac{SSR}{K-1}$ =0.917 | $F_R=\frac{MSR}{MSE}$ =1.231 | 4.76 |
| Between treatment | $SSK$ =4.25 | 3 | $MSK=\frac{SSK}{K-1}$ =1.42 | $F_K=\frac{MSK}{MSE}$ =2.119 | 4.76 |
| Error | $SSE$ =4 | 6 | MSE=0.67 | | |

| | |
|---|---|
| | Conclusion: Row, column and Treatment-varieties are accepted. (2M) |

7

**The following table shows the lives in hours of four brands of electric lamps.**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Brand A** | **1610** | **1610** | **1650** | **1680** | **1700** | **1720** | **1720** | **1800** |
| **B** | **1580** | **1640** | **1640** | **1700** | **1750** | | | |
| **C** | **1460** | **1550** | **1600** | **1620** | **1640** | **1660** | **1740** | **1820** |
| **D** | **1510** | **1520** | **1530** | **1570** | **1600** | **1680** | | |

**Perform an analysis of variance test the homogeneity of the mean lives of the four brands of lanms.**
**[A.U. A/M 2008, A.U N/D 2011, A.U Tvil M/J, A.U A/M 2015]**
**Answer : Dr.Balaji [P&S] Edition 1, page no : 4.14.**
**Step 1: N=26**
**Step 2: T=98**
**Step 3: C.F=369.39**
**Step 4: TSS=1950.61**
**Step 5: SSC=452.25**
**SSE=1498.36**
**Step 6: ANOVA TABLE**
**Step 7: Conclusion: accept**

8

**An experiment was designed to study the performance of 4 different detergents for cleaning fuel injectors. The following cleanliness readings were obtained with specially designed equipment for 12 tanks of gas distributed over 3 different models of engines.**

| | Engine 1 | Engine 2 | Engine 3 | Total |
|---|---|---|---|---|
| **Detergent A** | 45 | 43 | 51 | 139 |
| **Detergent B** | 47 | 46 | 52 | 145 |
| **Detergent C** | 48 | 50 | 55 | 153 |
| **Detergent D** | 42 | 37 | 49 | 128 |
| **Total** | 182 | 176 | 207 | 565 |

**Perform the ANOVA and test at 0.01 level of significance, whether there are differences in the detergents or in the engines.**
**[A.U. A/M 2008, A.U N/D 2011, A.U Tvil M/J, A.U A/M 2015]**
**Answer : Dr.Balaji [P&S] Edition 1, page no : 4.23.**
**Step 1: N=12**
**Step 2: T=-35**
**Step 3: C.F=102.08**
**Step 4: TSS=264.92**
**Step 5: SSC=135.17**
**Step 6: SSR=110.91**
**SSE=18.84**
**Step 7: Table of analysis of variance**
**Step 8: Conclusion: Reject**

9

**Test whether the five men differ with respect to mean productivity and test whether the mean productivity is the same for the four different machine types.**

| Workers | Machine Type | | | | Total | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | | $X_1^2$ | $X_2^2$ | $X_3^2$ | $X_4^2$ |
| $Y_1$ | 4 | -2 | 7 | -4 | 5 | 16 | 4 | 49 | 16 |
| $Y_2$ | 6 | 0 | 12 | 3 | 21 | 36 | 0 | 144 | 9 |
| $Y_3$ | -6 | -4 | 4 | -8 | -14 | 36 | 16 | 16 | 64 |
| $Y_4$ | 3 | -2 | 6 | -7 | 0 | 9 | 4 | 36 | 49 |
| $Y_5$ | -2 | 2 | 9 | -1 | 8 | 4 | 4 | 81 | 1 |
| Total | 5 | -6 | 38 | -17 | T=20 | 101 | 28 | 326 | 139 |

**[A.U. A/M 2008, A.U N/D 2011, A.U Tvil M/J, A.U A/M 2015]**

**Answer : Dr.Balaji [P&S] Edition 1, page no : 4.42.**

**Step 1: N=20**

**Step 2: T=20**

**Step 3: C.F=20**

**Step 4: TSS=574**

**Step 5: SSC=338.8**

**Step 6: SSR=161.5**

**SSE=73.7**

**Step 7: Table of analysis of variance**

**Step 8: Conclusion: Reject**

10

**The following is a latin square of a design, when 4 varieties of seeds are being tested. Set up the analysis of variance table and state your conclusion. You may carry out suitable change of origin and scale.**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| A | 105 | B | 95 | C | 125 | D | 115 |
| C | 115 | D | 125 | A | 105 | B | 105 |
| D | 115 | C | 95 | B | 105 | A | 115 |
| B | 95 | A | 135 | D | 95 | C | 115 |

**[A.U. A/M 2008, A.U N/D 2011, A.U Tvil M/J, A.U A/M 2015]**

**Answer : Dr.Balaji [P&S] Edition 1, page no : 4.53.**

**Step 1: N=16**

**Step 2: T=32**

**Step 3: C.F=64**

**Step 4: TSS=88**

**Step 5: SSC=4**

**Step 6: SSR=2**

**SSK: 22**

**SSE=60**

**Step 7: Table of analysis of variance**

**Step 8: Conclusion:**